
The Algorithm Tango: Voting Dynamics of “Dancing with the Stars”

Dancing with the Stars (DWTS) is a television show that combines the judges’ score and the fans’ vote by using the rank combining method and the percentage combining method to determine the winner. However, the potential divergence between judging criteria and fans preferences raises critical questions about the fairness of the show. To deal with this issue, this study develops a model to estimate fans’ voting behavior, evaluate the effects of the two combining methods, identify key factors influencing scores and votes, and propose a more balanced and fair competition system for future seasons.

First, after **data preprocessing** and **feature engineering**, we modeled fan votes through multiple linear regression, assuming a linear relationship between features and votes. The result estimation follows a **lognormal distribution** and was simulated using the **Monte Carlo Model**. The simulated voting was standardized and scaled by **scaling factor (SCF)**. The consistency rate between the output of this model and the actual data is **69.7%** by comparing the eliminated candidates. The certainty of this model could be revealed by the Coefficient Variation of **20.9%** and the Confidence Interval width ratio of **82.1%**.

Second, this study conducts a comparative analysis of rank method and percentage method for combining judge scores and fan votes throughout whole seasons. It shows **78.5%** eliminate results are the same according to both methods. **Spearman’s correlation** analysis confirms both methods have strong influence by fan votes. The average correlation for rank method is nearly **-0.63**, which is slightly greater than that of the Percentage Method of **-0.59**. The analysis of four controversial contestants illustrates two different methods will usually lead to the same result. By involving the Judges’ Saving mechanism, it will significantly prevent the controversial contestants getting high rank even if they have a low judge score. Therefore, the study recommends adopting the percentage-based method couples with the Judges’ Save rule.

The third task applies **multiple linear regression model** to analyze the factors affecting judge scores and fan votes. After data processing and **feature construction**, **ordinary least squares (OLS)** is used to estimate parameters. The result shows that the judge score model has a certain degree of explanatory power with $R^2 = 0.3045$, while the fan vote model ($R^2 = 0.1269$) is weaker due to the **fan “hit effect”**. Partner historical score, week progress and industry background have relatively higher absolute coefficient values, significantly impacting scores, clarifying key drivers of competition evaluations.

Fourth, the suggested system is a weighted percentage model with judge-save elimination. Judges’ weight increases linearly from **0.6** to **0.8** over the season, while fans’ weight decreases from **0.4** to **0.2**. The competition will be fairer and remain exciting for the fans. It also preserves fan impact early while ensuring technical merit dominates the finale. The suggested system for the four controversial contestants further illustrates its effectiveness and practicality.

Keywords: Monte Carlo Simulation; Multiple Linear Regression; Spearman’s Correlation

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1 Introduction

1.1 Problem Background

Dancing with the Stars is a popular American TV show where celebrities partner with professional ballroom dancers to perform themed, choreographed dances, which are then evaluated by expert judges [1]. This type of TV show originated in Britain as Strictly Come Dancing in May 2004, with the US adaptation premiering on ABC in June 2005. Since its launch, the successful show has been broadcast in at least 25 countries [2]. In this contest, we are focused on the 34 seasons of the US version. The judge from this contest can rate the couple from 1 to 10, and the fans can only give their vote online to the couple they like, not the couple they dislike.

Through the 34 seasons, the show has employed two distinct methods. The first method is to combine judge scores and fans' votes by ranking (Seasons 1, 2, and 28–34), which is judges' scores and fan votes were independently ranked, and the ranks were summed to determine elimination. The second method combines by percentage (Seasons 3–27), where raw judge scores and fan votes were converted to percentages of their respective totals and summed. These two methods respond to controversies where contestants with strong fan support, such as Jerry Rice in Season 2 is the champion despite lower judge scores than the other 3 contestants [3]. This shows the fans' vote can significantly influence the outcome of the competition.



[4] Figure 1. Dancing with the stars poster

1.2 Restatement of the Problem

Based on background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- Problem 1

Construct mathematical models to calculate the weekly total fan vote for each competitor throughout all seasons, evaluate how well these estimates match the real elimination results, and quantify the uncertainty surrounding the estimates.

- Problem 2

1. Compare how the result different in the method of combining judge and fan votes by rank and combining by percent. If the results are different, analyze which method has more benefit for the fans' influence.

2. Examining their impact on the cases in which the judges and the fans have different perspectives on the voting.

3. Give reasons for recommending the method for future seasons, according to the previous analysis, and indicate whether use the judges' selection of the worst two couples method.

- Problem 3

Develop a model to analyze how the various factors such as age and industry influence the judges' scores, and estimated fan votes as well as whether these effects are comparable.

- Problem 4

Provide an alternative approach that is fair based on weekly judge ratings and fan votes and give reasons.

1.3 Our Work

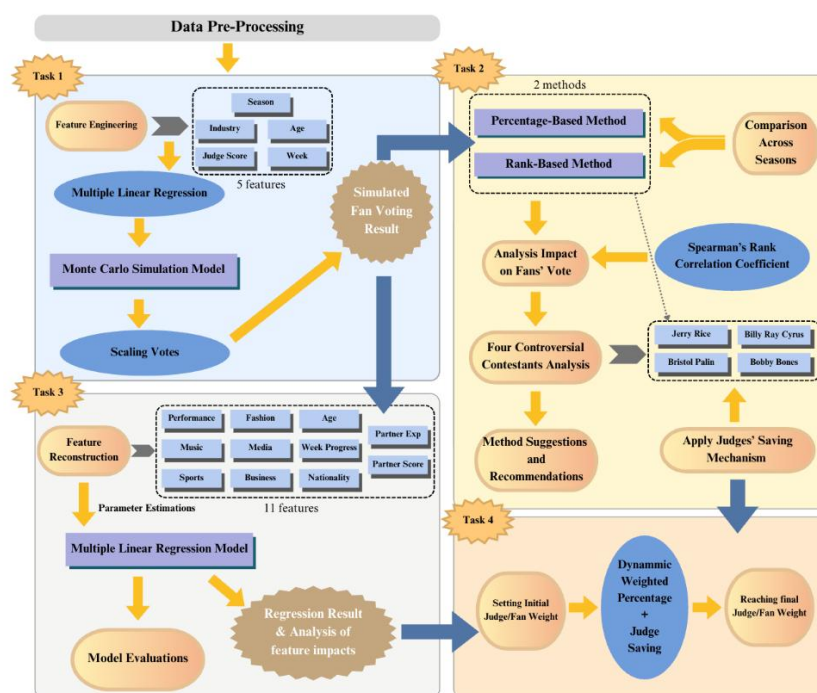


Figure 2. Our work overview

2 Assumptions and Justifications

By sufficient analysis of the problem, to simplify the models, the following well-justified assumptions are made:

1. Fan vote counts follow a log-normal distribution. Because fan votes exhibit the characteristics of non-negativity and right-skewedness, both of which are consistent with the properties of a log-normal distribution, and logarithmic transformation can convert the right-skewed data into a normal distribution to facilitate subsequent modeling.

2. When constructing the pseudo-target variable for fan votes, the weight of standardized judge scores (70%) is higher than that of other features (30%). As professionals, judges have important references in their scores.

3. Using 10% of the 9.24 million viewers (reported on November 26, 2025) as the estimated average weekly votes (100,000) to calculate the scaling factor (7047.63) is reasonable. This ensures simulated vote counts match the real-world magnitude, avoiding unrealistic results.
4. Fans can vote multiple times without limitation each week. Because it's hard to find every week's official rules and quantify the influence.
5. Core factors of celebrities and pro-dancers have a linear relationship with standardized judge scores and fan votes.
6. Standardizing continuous variables and mean-centering categorical variables can eliminate dimensional differences to ensure the coefficients are comparable.

3 Notations

Symbol	Description
$v_i(t)$	Simulated result of each fan voting
$\mu_i(t)$	Expected fan voting result based on feature vectors
β_0	Estimated Intercept
$\hat{\beta}_i; i = 1, 2, 3, 4, 5$	OLS estimators
SCF	Scaling factor on simulated fan voting results
$\bar{v}_i(t)$	Current statistical average votes
$C(\theta)$	Consistency rate
CV_i	Coefficient Variation
CI_i	Confidence Interval
R_i^S	Rank of contestant i based on judges' scores
R_i^V	Rank of contestant i based on fan votes
T_i^R	Total rank score for contestant i by using Rank Method
P_i^S	Percentage share of judges' scores for contestant i
P_i^V	Percentage share of fan votes for contestant i
T_i^P	Total percentage score for contestant i by using Percent Method
S_i	Judges' score for contestant i
V_i	Fan votes for contestant i
ρ	Spearman rank correlation coefficient
ρ_{FP}	Correlation between fan ranking and final ranking (Percent Method)
ρ_{FR}	Correlation between fan ranking and final ranking (Rank Method)
Judge_Score_Std	Standardized judge score (transformed to mean=0, std=1)
Fan_Vote_Std	Standardized fan score (transformed to mean=0, std=1)
Feature Variables	<i>Note: All categorical variables below are mean-centered binary indicators (1=belongs to category, 0=otherwise) for sample i</i>
BG_i	Set of mean-centered background dummies: {Performance, Music, Sports, Fashion, Media, Business}
$USACitizen_i$	U.S. citizen (same as above)
Age_i	Standardized age of celebrity (sample i)
$PartnerExp_i$	Standardized pro dancer appearance count (sample i)
$PartnerScore_i$	Standardized pro dancer historical average score (sample i)
$WeekProgress_i$	Normalized competition week progress (0=start, 1=end; sample i)
Model Parameters	
θ_0	Intercept (judge score model)

$\theta_j(j=1,\dots,11)$	Regression coefficient (judge score model, j-th feature)
γ_0	Intercept (fan vote model)
$\gamma_j(j=1,\dots,11)$	Regression coefficient (fan vote model, j-th feature)
ε_i	Residual (judge score model, sample i)
δ_i	Residual (fan vote model, sample i)
<i>Week</i>	the index of the current week in the season (e.g., week = 1,2, ...)
<i>Weekmax(s)</i>	the total number of weeks in seasons
<i>Week'</i>	the normalized week progress, which increases linearly from 0 in the first week to 1 in the final week of season
$T_i(Week)$	the combined score for contestant i in the given week, used to rank contestants and determine the bottom group

4 Data Description

This set of "Dancing with the Stars" dataset provides the basic information, but there are still some challenges: the number of judges in different seasons and weeks is inconsistent, the judges' scores and player attribute data are missing, there are outliers in the scores, and the use of complex wide-format storage methods is not suitable for direct analysis. To overcome these difficulties, we have carried out data cleaning to ensure the fairness of the comparison under configuration by different judges.

In the abnormal value detection, we used the range of mean ± 3 standard deviations to identify extreme scores. Instead of directly deleting these values, we used the clipping method to limit the abnormal values to the set of mean ± 3 standard deviation boundaries, while retaining the overall data distribution characteristics.

Before filling out the missing values of judge scores, we first determined whether the judges should be "normally present" but had the missing value or they were absent. Those judges who scored more than 50% of contestants in a specific season and weekly combination, would be regarded as "normally present". For the missing scores of these normally present judges, we adopt the hierarchical interpolation method: priority is given to the average scores of other judges in the same week, followed by the average scores of the players in other weeks, and when other methods are not effective, the average of the whole season is finally used. For missing values of celebrity attributes, we used mode imputation for categorical variables like industry and location, and median imputation for continuous variables like age.

Then the fundamental part of our cleaning process was this weighted average scoring scheme to address the challenge of inconsistent judge numbers. Since the 3-judge scenario takes a large proportion, we decided to normalize all scores on an equivalent 3-judge basis. For non-standard judge counts, we used this formula to normalize:

$$\text{weighted_total} = \text{raw_total} \times \left(\frac{3}{\text{num_judges}} \right) \quad (1)$$

After implementing the weighted scoring scheme, we deleted records of eliminated contestants where all judge scores were zero to clean up the dataset.

Finally, the code generated the new dataset based on the original data and the data cleaning process with a long-format structure. It was ordered by season and week with detailed metadata. It also contains both raw and weighted scores for next step's comparison.

5 Task 1: Monte Carlo Simulation Estimating Fan Votes

5.1 Feature Engineering

The given data presents both quantitative and qualitative features. To enhance the quantitative nature of the investigation, feature engineering quantified the qualitative elements and the quantitative elements have also been normalized.

Specifically, the weighted average score and age have undergone directly normalization. Notwithstanding, it is concerned that the one-hot and label encoding strategy may mislead the model in distinguishing a sequential relationship among industries and ballroom partners. To address this potential issue, the average score of the industry acts as the proxy variable for "industry strength"; while historical average score of the dance partner acts as an indicator of ability. Conversely, for time variables (week, season) the absolute time data is converted into relative progress (between 0 and 1) by proportion. The final placement has been achieved through a process of reverse normalization into positive indicators. These transformed variables have been utilized to form the feature vector for subsequent computations.

Table 1: Feature Engineering Transformations

Original Variable	New Variable	Transformation Formula
$J_i^{(s,w)}$: Weighted_average_score	$J_{std,i}^{(s,w)}$: Judge_score_std	$J_{std,i}^{(s,w)} = \frac{J_i^{(s,w)} - \mu_J^{(s,w)}}{\sigma_J^{(s,w)}}$
a_i : Age	$a_{norm,i}$: Age_norm	$a_{norm,i} = \frac{a_i - \mu_a}{\sigma_a}$
g_i : Industry	$I_{score,i}$: Industry_score	$I_{score,i} = \mu_J^{(g(i))}$
	$I_{norm,i}$: Industry_score_norm	$I_{norm,i} = \frac{I_{score,i} - \mu_I}{\sigma_I}$
p_i : Ballroom_partner	$P_{score,i}$: Partner_score	$P_{score,i} = \mu_J^{(p(i))}$
	$P_{norm,i}$: Partner_score_norm	$P_{norm,i} = \frac{P_{score,i} - \mu_P}{\sigma_P}$

$w_i^{(s)}$: Week	$w_{\text{norm},i}^{(s)}$: Week_norm	$w_{\text{norm},i}^{(s)} = \frac{w_i^{(s)} - \min(w^{(s)})}{\max(w^{(s)}) - \min(w^{(s)})}$
s_i : Season	$s_{\text{norm},i}$: Season_norm	$s_{\text{norm},i} = \frac{s_i - \min(s)}{\max(s) - \min(s)}$
r_i : Final_placement	$r_{\text{norm},i}$: Placement_norm	$r_{\text{norm},i} = 1 - \frac{r_i}{\max(r)}$

5.2 Establishment of Monte Carlo Model

The phenomenon of fan voting is always positive and usually shows a right-skewed distribution. Furthermore, after logarithmic transformation, it becomes an additive relationship, which is suitable for linear modeling. Hence, it is assumed that the Monte Carlo Model follows lognormal distribution:

$$v_i(t) \sim \text{LogNormal}(\mu_i(t), \sigma^2) \quad (2)$$

Where $\mu_i(t)$ is computed by the multiple linear regression of the feature vectors:

$$\mu_i(t) = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \dots + \beta_p \cdot x_p \quad (3)$$

Hence the distribution is equivalent to a normal distribution:

$$\log(v_i(t)) \sim \text{Normal}(\mu_i(t), \sigma^2) \quad (4)$$

The basic structure of mathematical expression of fans voting is:

$$\log(v_i(t)) = \text{feature vectors} + \varepsilon \quad (5)$$

Since the number of real fan votes is unknown, its values have been set as a Pseudo-target variable, where:

$$y_{\text{pseudo}} = w_{\text{judge}} \cdot J + w_{\text{other}} \cdot \sum (\alpha_j \cdot F_j) + \varepsilon \quad (6)$$

In equation (6), J represents the vector of standardized judge scores and F denotes the matrix of other features (e.g. age, industries...). The weights w_{judge} and w_{other} are established at 70% and 30% respectively as the judge score is the primary factor that influencing the fans' voting considerations, while other factors contribute lesser to their voting choices.

The progress of computation follows the least-square estimator algorithm, which target is to minimize the residual sum of squares $S(\beta)$, which is:

$$S(\beta) = \|y - X\beta\|^2 \quad (7)$$

$$\frac{\partial S}{\partial \beta} = -2X^T y + 2X^T X\beta = 0$$

That achieves the normal function:

$$X^T X\beta = X^T y \quad (8)$$

The OLS estimators consider :Intercept, Judge_score_std, Age_norm, Industry_score_norm, Week_norm, as illustrated in Table X, matching each $\hat{\beta}$ values respectively. The final solution of the Ordinary Least squares (OLS) estimator is:

Table 2: Outcomes of the OLS estimators

$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_4$	$\hat{\beta}_5$
0.021	0.703	-0.052	0.182	0.084	0.015

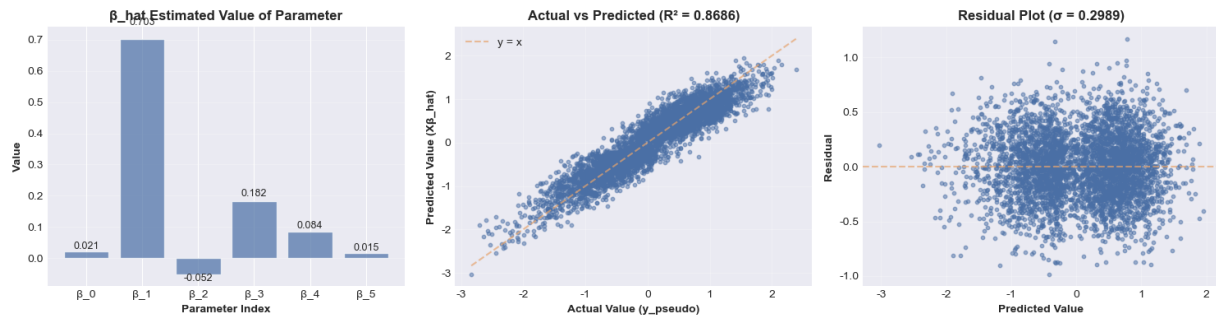


Figure 3: Estimated Value of Parameter

Figure 4: The Fitting Degree of Predicted Value and Actual Value

Figure 5: The Residual Plot for Predicted Value

Taking these estimators back to equation (3), the expectation of the normal distribution $\mu_i(t)$ can be computed. By repeating the simulation process for 500 times, set of logarithms of the fans voting has eventually been generated.

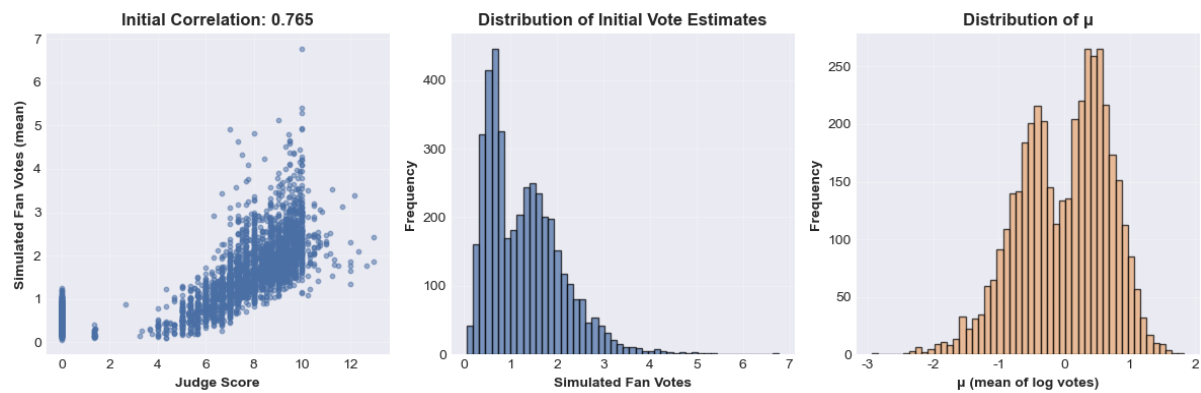


Figure 6. The Normal Distribution Series Simulated by Monte Carlo Model

However, due to the previous normalization of all the elements in the feature vectors, the model follows a standard normal distribution (as shown in Figure 6). To match with the real-world

data, a scaling factor is applied on the simulated fans voting results to conduct linear transformation with the formula below:

$$SCF = \frac{E[V_i(t)]}{\bar{v}_i(t)} \quad (9)$$

According to the news report about the DWTS show posted by November 26, 2025, the total number of the audience of the newest season of the show has reached 9.24 million. In this case, the estimated average voting has been set at about 10% of the total viewers for an approximation, which is 10,000. By computation, the scaling factor is 7047.63, the scaling operation has been conducted to the current dataset of $\mu i(t)$ by multiplying them together.

5.3 Computation of Consistency for Estimation

To measure the consistency with the real-world eliminated results in the show, constraint optimization has been utilized to enhance the model with higher practical ability.

The consistency rate is computed by:

$$C(\theta) = \frac{1}{T} \sum_{t=1}^T I\{Est(t) = Act(t)\} \quad (10)$$

Each week, the actual eliminator must satisfy the following constraint:

Table 3. Votes Calculation and Constraints Based on Rand and Percentage Method

	Rank Method	Percentage Method
Judge Votes	$R_{\text{judge}}(i) = \text{rank}(-s_i)$	$P_{\text{judge}}(i) = \frac{s_i}{\sum_j s_j}$
Fan Votes	$R_{\text{fan}}(i) = \text{rank}(v_i)$	$P_{\text{fan}}(i) = \frac{v_i}{\sum_j v_j}$
Total Votes	$R_{\text{total}}(i) = R_{\text{judge}}(i) + R_{\text{fan}}(i)$	$P_{\text{total}}(i) = P_{\text{judge}}(i) + P_{\text{fan}}(i)$
Constraint	$R_{\text{total}}(e) = \max(R_{\text{total}})$	$P_{\text{total}}(e) = \min(P_{\text{total}})$

Among the formulas in Table 3, the constraints are set for rank and percentage method where the estimated eliminator $\hat{e}_t$ has the largest ranking and the smallest percentage of total voting. (The detailed algorithm can be referred to Algorithm 1 in Appendix.)

To optimize the result, a further constraint has been added to the consistency computation:

$$v_e \leq \alpha \times \min_{i \in I_t} v_i \quad (11)$$

It was assumed that the $\alpha = 0.8$, which ensures that the votes of the eliminated candidates shall not exceed 80% of the minimum number of votes.

By operating Algorithm 1, the computed consistency rate of the Monte Carlo Model is $C(\theta) = 0.697$, suggesting a discrepancy of 31.3% between eliminated participants from the simulated and actual results. The elimination results of the show are not entirely determined by the judges' scores and fans' votes in accordance with the publicly available rules. Some other factors may be involved.

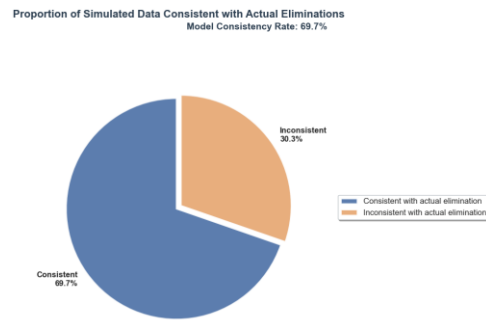


Figure 7. Proportion of Simulated Data Consistent with Actual Eliminations

5.4 Model Evaluation by Certainty Detection

Mathematically, two parameters have been utilized in certainty detection, Coefficient Variation CV_i and Confidence Interval CI_i , where:

$$CV_i = \frac{\sigma_{v_i}}{\mu_{v_i}} \times 100 \quad (12)$$

$$CI_i = [\mu_{v_i} - z \cdot \sigma_{v_i}, \mu_{v_i} + z \cdot \sigma_{v_i}] \quad (13)$$

Preliminary results indicate a **20.9%** probability of CV_i , suggesting a high degree of uncertainty in the outcomes, which is consistent with the observed pattern of eliminations on a weekly basis. This value suggests a series of potential factors: the model was established based on an inverse problem, as the actual fan voting result has not been published, the entire process is to optimize and simulate from the only attribute, which is the elimination result.

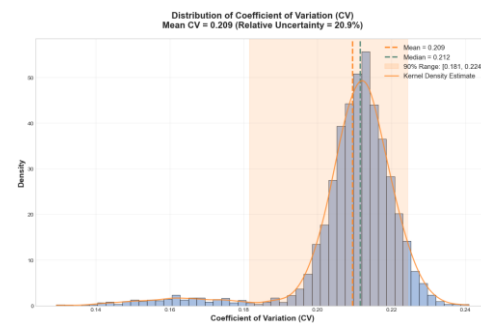
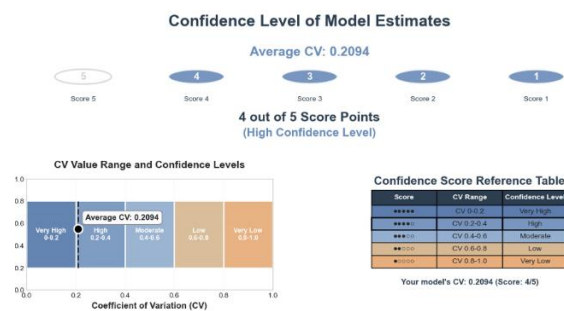


Figure 8. Visualization for Confidence Variation of Model Estimates

Figure 9. Distribution of Coefficient Variation (CV)

As shown in Figure 9, the coefficient of variation (CV) distribution of all simulated fan voting values is concentrated and symmetrical, with an average of 20.9% and a median of 21.2%. 90% of the CV values fall within the narrow range of 18.1% to 22.4% (see the orange block in the figure). This indicates that the relative variation levels among the simulated fan voting are highly consistent, and the process is in a stable and controlled state.

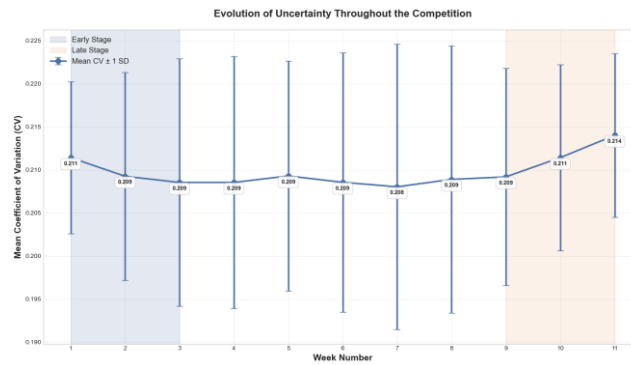


Figure 10. Evolution of Uncertainty Each Week Throughout the Competition

As illustrated in Figure 10, the trend of average weekly CV throughout each season of the competition is revealed. Since the number of weeks in each season of competition may vary, the average CV has been standardized to 11 weeks, representing the largest number of weeks among all the seasons. A thorough examination of this diagram reveals that the highest CV mean emerges at week 11 (the final week). Moreover, the diagram illustrates a low-side among high trends, where the CV is higher at the early and late stage, as the highlighted parts in the diagram.

Moreover, for CI_i , a 95% confidence level has been established for the confidence interval test, which the z-score value of $z = 1.96$. The result for CI_i is **82.1%**, indicating that the 95% confidence interval width is **0.821** times the average estimate value.

6 Comparative Analysis of Rank and Percent Methods for Combining Judge and Fan Votes

6.1 Method Definition and Mathematical Model

To assess the impact of different vote combination methods, both the Rank and Percent approaches apply to each season using estimated fan vote data together with judges' scores. The mathematical formula for each method is as follows:

Rank-Based Method

The Rank method assigns a rank to each contestant based on total judges' scores and total fan votes separately, then sums these ranks to determine elimination.

$$R_i^S = rank(S_i) \quad (14)$$

$$R_i^V = \text{rank}(V_i) \quad (15)$$

$$T_i^R = R_i^S + R_i^V \quad (16)$$

Percentage-Based Method

The Percent method calculates the percentage share of judges' scores and fan votes for each contestant, after that sum up these percentages to identify the contestant with the lowest combined score for elimination.

$$P_i^S = \frac{S_i}{\sum_{j=1}^n S_j} \quad (17)$$

$$P_i^V = \frac{V_i}{\sum_{j=1}^n V_j} \quad (18)$$

$$T_i^P = P_i^S + P_i^V \quad (19)$$

6.2 Comparison of Rank and Percent Methods Across Seasons

The following pie chart shows the percentage of weeks with the same and different elimination results between the two approaches.

There is a high degree of consistency when both systems are applied to all available seasons, which approximately 78.5% of weekly eliminations are the same under both methods. This demonstrates that method choice has little effect on the outcome of DWTS competition.

The 21.5% disagreement rate indicates that, even though both methods usually converge, their underlying mathematical foundations will produce different results in about one out of fifth elimination decisions. In minority cases, the selection of the combining method will impact directly on which competitors move on to the next round.

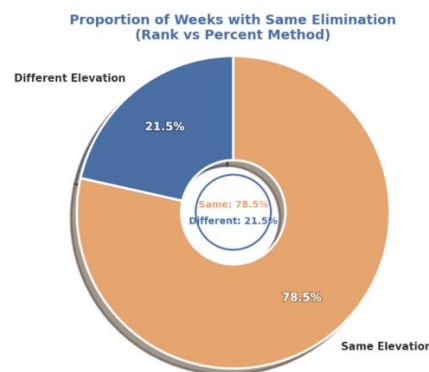


Figure 11. Proportion of Weeks with Same Elimination

6.3 Impact of fans' vote to two different methods

Spearman's Rank Correlation Coefficient

We use the Spearman rank correlation coefficient to evaluate the effect of fans' voting on the final ranks with both rank and percentage techniques. This nonparametric statistical technique is especially useful for ordinal data, because it can assess the monotonic connection between two ordinal variables. ρ is a number that range between -1 to 1, d_i is difference between the two ranks of each observation, n is number of observations [5].

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad (20)$$

Then, calculate ρ_{FP} which is correlation between fan ranking and final ranking under the percentage method and the ρ_{FR} which is correlation between fan ranking and final ranking under the rank method and compare ρ_{FP} and ρ_{FR} which one have more impact on fans' vote.

Result of Spearman rank correlation coefficient for two methods

Across all seasons, both approaches show high negative correlations. This imply that the number of the vote by fans have a big impact on final placements of the contestant.

Average Rank Method: -0.63

Average Percentage Method: -0.59

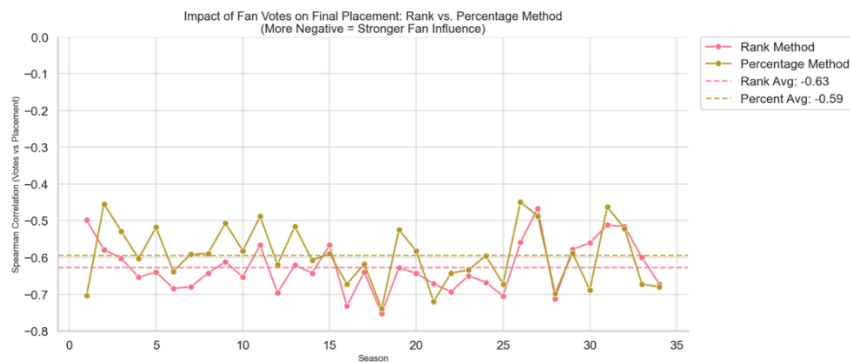


Figure 12. Impact of Fan Votes on Final Placement: Rank vs. Percentage Method

From Figure 12, fan influence is slightly greater when using the rank technique, with a correlation greater magnitude of 0.04. This suggests that compared to the percentage method, the rank method is slightly more sensitive to fan preferences. However, the difference between the two systems is generally tiny, showing that the choice between them does not significantly affect the balance of power between judges and fans, but it may determine certain elimination outcomes in close competitions.

6.4 Four Controversial Contestants Analysis

As Figure 13 illustrates, four contestants include Jerry Rice (Season 2), Billy Ray Cyrus (Season 4), Bristol Palin (Season 11), and Bobby Bones (Season 27) they all consistently received low scores from the judges but achieved high final placements because of the strong fans support. This emphasizes the significant disconnect between judges' scores and fans votes in DWTS competition, highlighting how public popularity directly influences competition outcomes. As a result, it contributes to the discussions about the reliability of the two combining

methods and the fairness of the voting system.

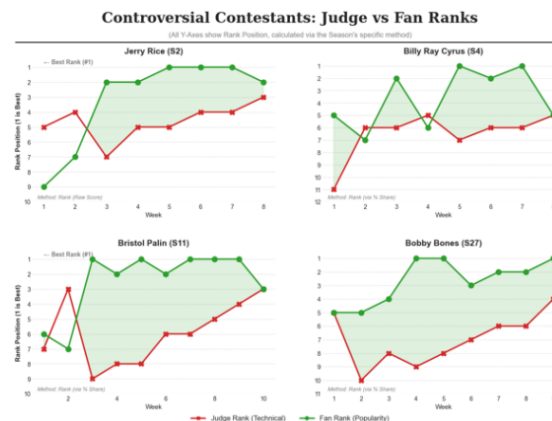


Figure 13. Controversial Contestants: Judge vs. Fan Ranks

Four Controversial Contestants by Using Two Different Method

It is important to analyze whether these two different approaches will affect their rankings and the outcome for these four typical controversial contestants.

According to the Jerry Rice, Billy Ray Cyrus, Bristol Palin the ranking method and percentage method showed little difference in their weekly rankings. But the final result for they are all the same by using two different methods. Form the three according graph it show Jerry be the 2nd place, Billy Ray Cyrus get 5th place (Eliminated in Week 8), Bristol Palin is the 3rd place. The first major reason for this result is these two methods have a high level of consistency rate, and another reason is both methods are sensitive to the fans' vote that cause controversy.

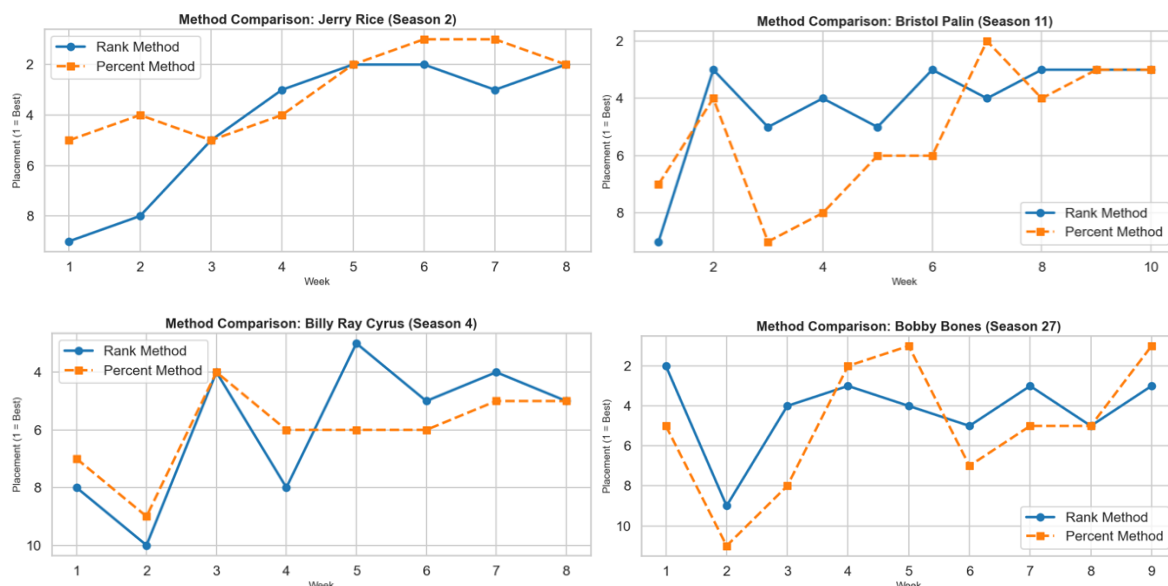


Figure 14. Method Comparison of the four controversial contestants

For the Bobby Bones case the choice of methods is matter. With the percentage combination method, he will win first place in this season, however when applying the rank base method,

he will be the 3rd place, because there is possibility that the calculation falls in the disagreement between the two methods.

Apply Judges' Saving Mechanism

The rule of this approach is contestants must total scores that rank the bottom two. And the judge has the choice eliminate one among two of them, which means the contestant with the lower judge score for that week will be eliminated.

This mechanism will make dramatic impact on the controversial contestants, because they often stay in the danger zone. Their total scores always combining low judge scores, and high fan votes. If become the last two the definitely have disadvantage, because they usually have to compete with the opponents who have higher judge scores (though possibly have fewer fan votes).

Figure 15 shows that Jerry Rice will be eliminate in week 7, which means he cannot advance to the finals. Similarly, Bristol Palin and Bobby Bones repeatedly triggered rules during the latter weeks, this will ultimately be preventing them from getting high rankings. Billy Ray Cyrus had already finished fifth in the historical standings, but applying this mechanism will let him to exit one to two weeks earlier or even eliminate at the beginning of that season.

This rule effectively correct extreme controversial cases, which prevent contestants with low judge scores but high fan's vote from advancing too far, especially into the finals or winning the championship.

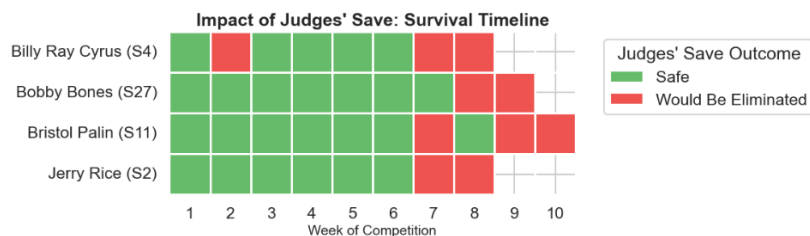


Figure 15. Impact of Judges' Save: Survival Timeline

Based on the analysis above, it is better to use percentage combining method with Judges' Save. Percentage combining method have less sensitivity to the fans' vote than the rank combining method, but it still have a great correlation with fans' vote. So, applying Judges' Save is necessary, this will let the controversial contestants be eliminated in advance, which will make the DWTS fairer.

7 Multiple Linear Regression Analysis of Factors Influencing Dance Competition Scores

7.1 Feature Construction

The first step is pulling the data from Task1 in and then checking the required columns. And

after standardizing the judges' scores and fans' votes, both distributions have been transform into the distribution with mean = 0 and standard deviation = 1 to prevent the influence of different dimensions.

$$y_{std} = \frac{y - \mu_y}{\sigma_y} \quad (21)$$

And then the age data is processed. This involves checking for the existence of the age column, and if missing, filling it with randomly generated integers between 18 and 70.

The next step is dividing the original industry information into six categories based on the key words: performance, music, sports, fashion, media and business for easier analysis. Here is the formula for creating virtual variables on account of the classification:

$$has_category_background_i = \begin{cases} 1 & \text{if sample } i \text{ belongs to this category} \\ 0 & \text{otherwise} \end{cases} \quad (22)$$

The fourth influencing factor is nationality, so we extracted nationality information and create a binary variable to determine whether one is a U.S. citizen.

Meanwhile, we constructed a week progress feature to find out whether the scoring scale will be influenced by week progress. The original week numbers were normalized to a [0,1] scale, where 0 represents the start of a season and 1 represents the end. Here is the normalizing formula:

$$week_progress = \frac{week - \min(week)}{\max(week) - \min(week)} \quad (23)$$

In addition to the influence of characteristics of the celebrities, the impact of various pro dancers also needs to be analyzed. So, the final step is calculating the two influence factors of pro dancers. The experience of pro dancers is simplified to the sum of the number of appearances. And their historical scores are regarded as the average score of all the matches they participated.

All continuous variables are standardized to ensure comparability across different scales:

$$age_{std} = \frac{age - \mu_{age}}{\sigma_{age}} \quad (24)$$

$$partner_exp_{std} = \frac{partner_exp_total - \mu_{partner_exp}}{\sigma_{partner_exp}} \quad (25)$$

$$partner_score_{std} = \frac{partner_historical_score - \mu_{partner_exp}}{\sigma_{partner_exp}} \quad (26)$$

$$week_progress_{std} = \frac{week_progress - \mu_{week_progress}}{\sigma_{week_progress}} \quad (27)$$

The final feature set includes 11 variables: 6 mean-centered professional background variables

(USA citizen, performance/music/sports/fashion/media/business background) and 4 standardized continuous variables (age, partner experience, partner historical score and week progress).

7.2 Multiple Linear Regression Model Construction

Model Specification and Mathematical Formulation

Since mean-centered multiple linear regression model can offer clear coefficient estimates that directly interpret the magnitude and direction of each factor, it is beneficial for estimating all features' impact of judge scores and fan votes. And the model specification is:

Judge Score Model:

$$\begin{aligned} \text{Judge_Score_Std} = & \theta_0 + \theta_1 \times \text{Performance}_i + \theta_2 \times \text{Music}_i + \theta_3 \times \text{Sports}_i \\ & + \theta_4 \times \text{Fashion}_i + \theta_5 \times \text{Media}_i + \theta_6 \times \text{Business}_i + \theta_7 \times \text{USACitizen}_i \\ & + \theta_8 \times \text{Age}_i + \theta_9 \times \text{PartnerExp}_i + \theta_{10} \times \text{PartnerScore}_i \\ & + \theta_{11} \times \text{WeekProgress}_i + \varepsilon_i \end{aligned} \quad (28)$$

Fan Vote Model:

$$\begin{aligned} \text{Fan_Vote_Std} = & \gamma_0 + \gamma_1 \times \text{Performance}_i + \gamma_2 \times \text{Music}_i + \gamma_3 \times \text{Sports}_i + \gamma_4 \times \text{Fashion}_i \\ & + \gamma_5 \times \text{Media}_i + \gamma_6 \times \text{Business}_i + \gamma_7 \times \text{USACitizen}_i + \gamma_8 \times \text{Age}_i \\ & + \gamma_9 \times \text{PartnerExp}_i + \gamma_{10} \times \text{PartnerScore}_i + \gamma_{11} \times \text{WeekProgress}_i + \delta_i \end{aligned} \quad (29)$$

Parameter estimation methods

OLS (Ordinary Least Squares) is the method to estimate the parameters of the two regression models. The core principle of this method is to find a set of regression coefficients β to minimize the total discrepancy between the model's predicted values $\hat{y}_i$ and the observed values y_i . Here is the formula:

$$\min_{\beta} S(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 \quad (29)$$

After taking the derivative of the previous formula and setting it to zero, we can find out:

$$\hat{\beta} = (X^T X)^{-1} X^T y \quad (30)$$

Where X is the design matrix including an intercept column, and y is the vector of response variables. To handle potential multicollinearity issues, a pseudoinverse approach is implemented when the design matrix is singular:

$$\hat{\beta} = X^+ y \quad (31)$$

Where X^+ denotes the Moore-Penrose pseudoinverse of X . This robust estimation approach ensures stable coefficient estimates even in the presence of multicollinearity.

7.3 Regression Result Analysis and Interpretation

Coefficient Significance Analysis

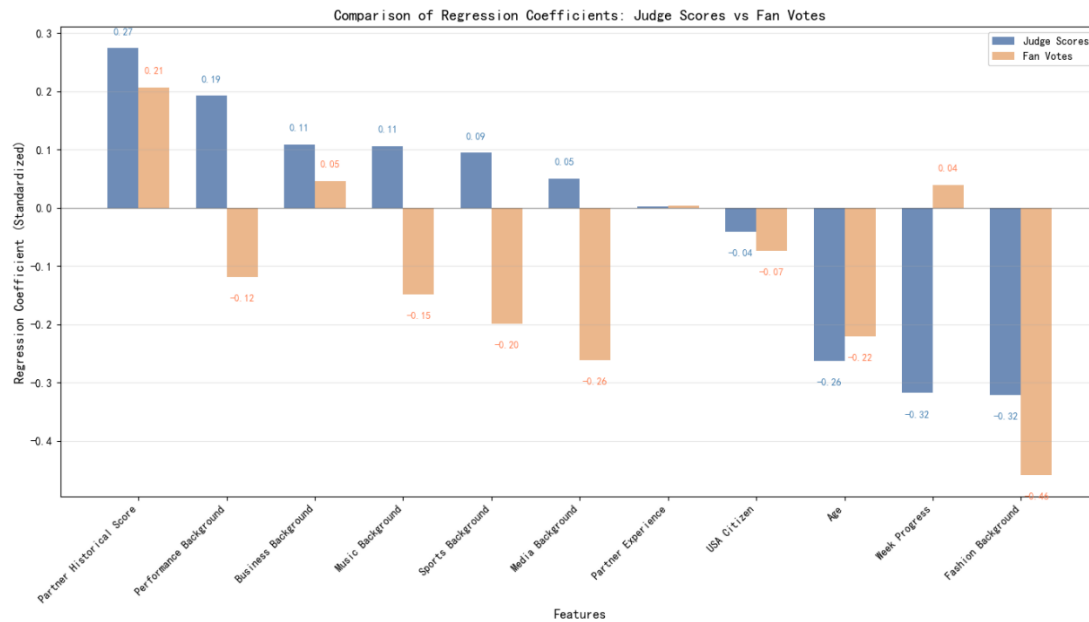


Figure 16. Comparison of Regression Coefficients: Judge Score vs. Fan Votes

Table4: Regression coefficient results

Feature	Judge Coef	Fan Coef	Difference
Performance Back-ground	0.192	-0.119	0.311
Music Background	0.106	-0.149	0.255
Sports Background	0.094	-0.199	0.294
Fashion Background	-0.321	-0.459	0.138
Media Background	0.050	-0.262	0.312
Business Back-ground	0.109	0.046	0.063
USA Citizen	-0.042	-0.073	0.032
Age	-0.263	-0.220	0.043
Partner Experience	0.002	0.003	0.001
Partner Historical Score	0.274	0.207	0.067
Week Progress	-0.318	0.040	0.358

Note: Since the coefficients of partner experience are so small, the analysis will ignore this variable of both judge scores and fan votes regression.

Firstly, it's obvious that the coefficient of Week Progress is large and statistically significant, indicating a strong effect on Judge scores. While it also has the largest divergence between judges and fans, only has slight effects on fan votes.

Then the difference of performance, music, sports and media show the esthetic differences between judges and fans. And these are also influential factors largely driven by personal tastes and preferences.

Fashion Background is the most negative factor of judge scores and fan votes. This may cause from public's opinion that these people don't have enough dance experience or focus more on fashion outfits. Moreover, elderly contestants and USA citizens may lead to decrease of judge scores and fan votes. Judges and fans may have more interests and curiosity about celebrities from other countries besides USA, but this situation just slightly influences the result.

Partner Historical Score is one of the most positive factors with 0.274 judge coefficient and 0.207 fan coefficient, which means a professional dance partner plays a fundamental role in this competition. Meanwhile, celebrities with business background may get better scores and votes due to their special industries.

Key Feature Impact Analysis

1) Age Effect

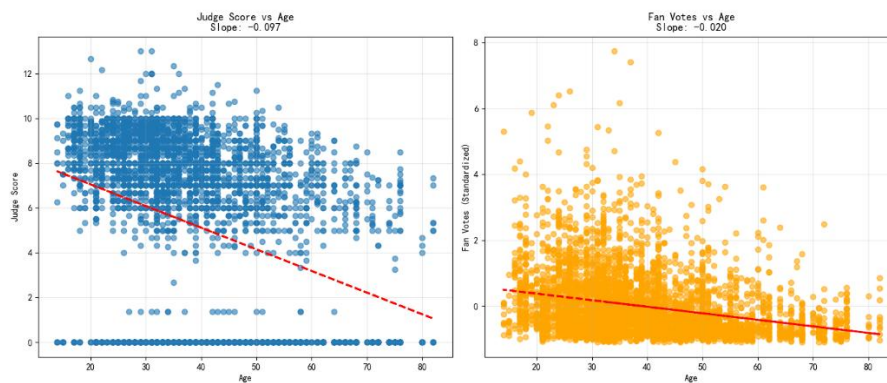


Figure 17. Judge/Fan Score vs. Age Slope

From Figure 17, it shows that age affects both judge score and fan votes negatively. That also indicates the dilemma which elderly celebrities faced---people, especially the judges, may have negative impression of this group due to their age and physiological disadvantages. This is also a reflection of the constancy and possible injustice that the elderly may encounter in their daily lives. (The line between 0-2 of the figure on the left is due to filling out the 'N/A' missing values in data preprocessing step.)

2) Professional Background Effect

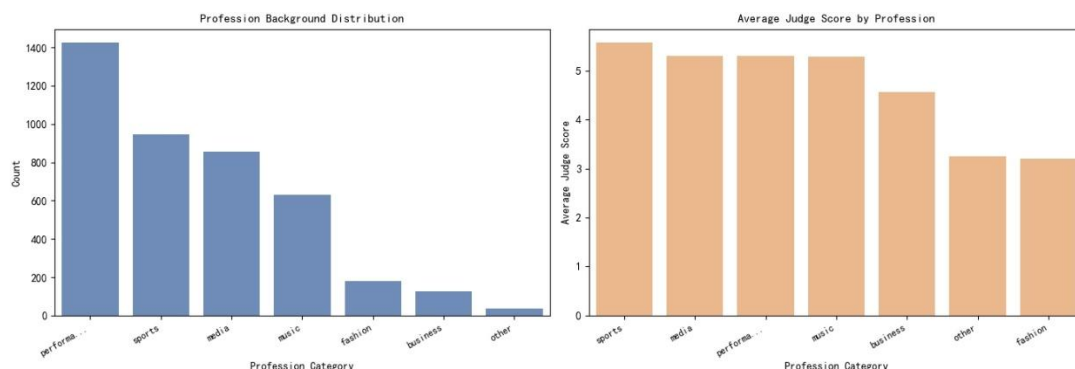


Figure 18. Profession Background Distribution & Average Judge Score by Profession

The left bar chart shows that performers make up the largest group in the dataset, while the number of business professionals are low. The right chart reveals that judges prefer giving higher average scores to celebrities from sports, media, performance and music categories rather than fashion and other categories.

3) Competition Progress Effect

In figure 19, judge scores show a significant downward trend as the week progresses (slope = -3.899), while fan votes show a slight increase over the same period (slope = 0.124).

This shows that as the week progresses, the judges' requirements become more stringent, so the average score keeps decreasing.

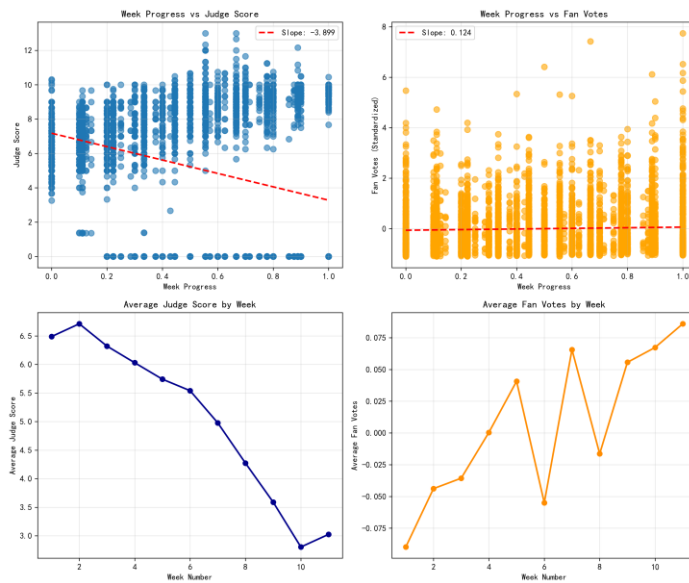


Figure 19. Week Progress Analysis

7.4 Model Evaluation and Further Discussion

Model Results and Evaluation

Judge Score Model:

$$R^2: 0.3045$$

$$\text{Adjusted } R^2: 0.3027$$

Fan Vote Model:

$$R^2: 0.1269$$

$$\text{Adjusted } R^2: 0.1246$$

R^2 (Coefficient of Determination) measures the proportion of variance explained by model. It is closer to 1, the fit is better. And $\text{Adjusted } R^2$ penalizes excessive features to better reflect the explanatory power of model.

From the result, we can find that the judge score model has some explanatory power, while the fan vote model exhibits relatively weak explanatory power. So these existing factors are not enough to estimate the behaviors and preferences of fans.

To further evaluate the fitting quality of the model, we conducted residual diagnostic analysis on the two regression models, and here are the visualizations:

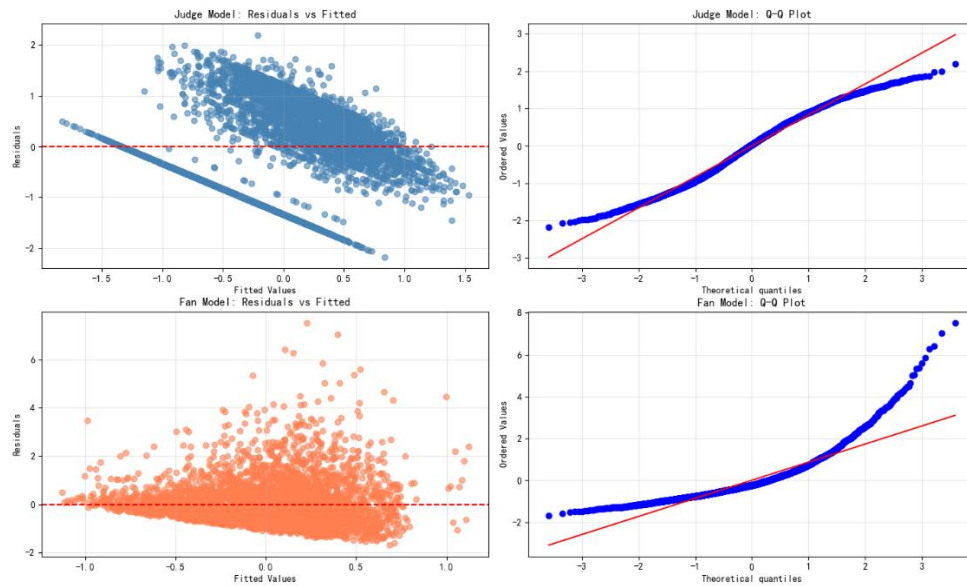


Figure 20. Residual Plot & Q-Q plot

Note: The horizontal axis represents standardized fitted values.

For the Judge Scores model, the trend line of the upper left scatter plot shows a downward slope, indicating that the residuals decrease with the increase of fitted values. Therefore, there exists mild heteroscedasticity. In the Q-Q plot (top right), the residual points are slightly above the normal reference line at the left tail and below the line at the right tail, indicating that judges are stricter to outstanding celebrities and tolerant to relatively weaker celebrities. And this is a common phenomenon in subjective scoring data, and it does not seriously violate the normal distribution assumption. So the model's residual characteristics are relatively stable.

For the Fan Votes Model, the scatter plot on the bottom left shows that the degree of dispersion of the residuals increases significantly with the increase of the fitted value. And in the Q-Q plot of fan model, the left tail is slightly below the reference line, while the right tail is significantly above the reference line. Therefore, for the residual points of popular celebrities do not fit the normal distribution. Both figures show that the fan votes of popular contestants have many extreme values due to the influence of social media communication and other factors. Also, it reflects the fan “hit effect” in the current internet era.

In summary, the diagnostic results of the two models are consistent with the explanatory power characteristics reflected by R^2 . The judge scores model has a certain degree of explanatory power and captures the core influential factors of scores. Although there is a more obvious heteroscedasticity and deviation from the normal distribution, it objectively reflects the high heterogeneity of fan voting behavior, and the impact of the core variables identified by the model is still statistically significant.

Model Strength

The linear regression model clearly showcases the relationship between influential factors and

judge scores / fan votes with strong result interpretability. The coefficients can directly quantify the impact of each core variable on judge scores and fan votes. Moreover, standardization and centralization are applied to eliminate the influence of different dimensions on regression results. Multi-dimensional model verification is completed by combining fitting degree tests and residual diagnosis, with reliable and non-one-sided conclusions. The model framework is simple and highly expandable, which can be easily optimized with more data or combine new influence factors in the future.

Model Weakness and further improvement

Also, there are still some limitations of models. The first one is lacking variable dimensions and data resources. We will take digital indicators such as topic popularity, fan volume and interactions on social media platforms into the model and combine questionnaire surveys to solve this problem and enhance the model's explanatory power. Another limitation is the model misses some nonlinear effects in analysis. Methods such as random forests and LASSO regression capture nonlinear variable interactions, and highlight the key drivers of judge scores and fan votes through their feature importance ranking functions.

8 A Fairer System Based on Judges' Scores and Fan Votes

8.1 Design Objectives and Fairness Principles

The result from the previous task shows both the rank and percentage methods are affected by the fans' vote and the percentage method is less sensitive than the rank method. Hence, in the new model, it is better to adjust the weighting applied to judges and fans through the percentage combining method. Meanwhile, the Judges' Save mechanism used in some seasons can fix extreme cases, so we propose a new system that use weighted percentage method along with judge save elimination.

Initially, we set the weight for the judge at the beginning of the seasons is 0.6 and 0.4 for the fans' vote. The reason for that is the judges' weighting must always maintain a slight advantage, as the original percentage method would be influenced by fan voting. However, the judges' weighting was not set too high initially because the influence of fans must be clearly visible, especially in the early stages of the DWTS TV show. Otherwise, the show would become a judges' base competition, which would lead to a decline in audience engagement and less exciting for fans. In the last week of the season the weight of the Judge should set to 0.8, the fan vote weight will be 0.2 because the judge's evaluation becomes dominant, which can ensure the champion has sufficient technical merit instead of popularity.

$$T_{initial} = 0.6P_i^S + 0.4P_i^V \quad (32)$$

$$T_{end} = 0.8P_i^S + 0.2P_i^V \quad (33)$$

To allow weights to change over time, we introduce $Week'$, which will make the judges' and fans' weights a linear function of week progress. So, the final weighted percentage model that take fairness and excitement into consider is $T_i(week)$.

$$Week' = \frac{week - 1}{Week_{max}^{(s)} - 1} \quad (34)$$

$$T_i(Week) = (0.6 + 0.2 Week')P_i^S + (0.4 - 0.2 Week')P_i^V \quad (35)$$

8.2 Verify the new system Through Four Controversial Contestants

By applying the new system of weighted percentage with the judge save elimination can give for the controversial cases the result will make the competition fairer and give fans a sense of involvement. Figure 21 demonstrates that the new system has pushed them to be eliminated earlier and have lower final placements, in line with their lower technical performance. Importantly, they are not eliminated at the very early stage, which shows that audience engagement is still preserved.

Contestant	Season	Historical Final Place	New Rule Final Place	Historical Elimination Week	New Rule Elimination Week
Jerry Rice	2	2	4	reach final	Week 7
Billy Ray Cyrus	4	5	7	Week 8	Week 5
Bristol Palin	11	3	9	Reaches Final	Week 4
Bobby Bones	27	1	8	Champion	Week 6

Figure 21: Historical versus simulated results under the proposed rule (1=champion)

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MEMORANDUM

✧ Dancing with the Stars

To: DWTS Producers
From: MCM Modeling Team
#2611291
Date: February 2, 2026

Dear Sir or Madam,

We are pleased to provide strategic assistance for planning the show "Dancing with The Stars" in the upcoming seasons. Based on the given data, we have developed mathematical models in demonstrating the factors that affect the fan voting, and made comparisons between different voting rules during the past seasons. Finally, we have developed a more practical and scientific voting system. Our insights, approaches and suggestions are illustrated below:

MEMO

Key insights:

- There is a **high consistency** in both voting methods (percentage and rank method) with a consistency rate at approximately **78.5%** on weekly elimination results, showing little difference in the outcomes.
- By Spearman's rank correlation coefficient, the average coefficient on rank method is **-0.63**, while for percentage method the average is **-0.59**. As the more negative the coefficient indicates a stronger fan impact in contributions to the voting results, rank method shows a **higher** incidence.
- Based on the given four extreme cases (Jerry Rice, Bristol Palin, Billy Ray Cyrus, Bobby Bones), the data reveals that though they continuously received **low** judge scores for **consecutive weeks**, they still obtained good final results eventually, which is caused by their massive fan support. We observed that for Jerry Rice, Billy Ray Cyrus, Bristol Palin, the rankings when applying two methods are all the same. However, for Bobby Bones' case, the ranking has **dropped by 2** when switching from percentage method to rank method.
- When applying the **judge's saving mechanism** that appeared in **Season28**, the controversial contestants may encounter significant risk in being eliminated as they usually have to compete with the opponents who have higher judge scores. It has been illustrated in the data that the application of the judge's saving method would eliminate the controversial contestants at very early stages.
- From the sensitive analysis on the parameters of the multiple linear regression model, it was observed that candidates' lable of fashion industry would have put the most negative influence in their score with a coefficient of **-0.318**. Also, the later the weeks of the show goes by, the **stricter** score would be given by the judges.

Based on our insights, we may suggest the following alternatives on voting rules in the following seasons :

- Use percentage method along with the judge's saving strategy. Set the weight for the judge at the beginning of the seasons is **0.6** and **0.4** weighted for the fans' vote at the beginning of the season.
- The weight of judge and fan would run dynamically. The judge weight would ascend while the weight of fan would decline throughout the season at a constant rate named as **Week'**, computed by the total number of weeks of a season.
- In the last week of the season, the weight of the should be **0.8**, fans vote weight will be **0.2**.

We sincerely wish that our proposal would be helpful in the future program plannings for the show. If you have any enquiries about any suggestion, please feel free to contact us.

Sincerely,
MCM Modeling Team#2611291

Appendix

Algorithm 1

```

for each week  $t$  :
    extract this week's data:  $S_t, \bar{V}_t, e_t$ 
    if (Rank Method):
         $\hat{e}_t = \max(R_{\text{total}})$ 
    elif (Percentage Method):
         $\hat{e}_t = \min(P_{\text{total}})$ 
    if ( $\hat{e}_t = e_t$ ):
        return True
    else
        return False

```

Report on Use of AI Tools

DeepSeek (2026 Version)

Query1: Based on our task to estimate unknown fan votes from elimination results, we've decided to use a Monte Carlo simulation model. What specific probability distribution should we assume for the fan votes, and how should we parameterize it to incorporate judge scores and contestant features?

Output:

Adopted Model Specification:

We will assume that the **logarithm of fan votes** for each contestant follows a **Normal distribution**. This is the **log-normal model** that was ultimately adopted in our implementation.

Parameterization:

The mean parameter $\mu_i(t)$ for contestant i in week t is modeled as a linear combination of:

1. **Standardized judge scores ($J_i(t)$):** Using `weighted_average_score` standardized by season and week to remove scale effects.
2. **Contestant feature vector (X_i):** This includes encoded categorical variables (top industries, age groups) and other relevant features like `season_experience`.
3. **Week fixed effects (γ_t):** To capture weekly variations in overall voting enthusiasm or show viewership.

The adopted generative formula is:

text

$$\log(v_i(t)) \sim N(\mu_i(t), \sigma^2)$$

$$\mu_i(t) = \alpha * J_i(t) + \beta * X_i + \gamma_t$$

Why This Was Adopted:

- It ensures positive vote estimates.
- It naturally integrates our predictors (judge scores, features) into the core of the simulation.
- The log-normal distribution can model the right-skew often expected in popularity metrics.

Query2: brain storming the systems using fan votes and judge scores each week that you believe is more “fair”

Output:

1. Weighted Hybrid Scoring System

Idea: Assign dynamic weights to judge scores and fan votes based on competition stage.

Example:

- Early weeks: 40% judges, 60% fans (encourage audience engagement).
- Semifinals: 50% judges, 50% fans (balance).
- Finals: 60% judges, 40% fans (emphasize technical skill).

Fairness rationale: Recognizes both expertise and popularity while adjusting influence as stakes increase.

2. Rank-Weighted Borda Count

Idea: Use a Borda count system where each judge's ranking and fan ranking contribute points.

Example: If there are n contestants:

- Judges' average rank gives points: $n - \text{rank}$.
- Fan votes converted to ranks, same scheme.
- Sum points from both sources.

Fairness rationale: Reduces impact of outliers and makes each ranking position matter equally.

3. Proportional Influence with Safety Threshold

Idea: Combine percent-based scoring but add a "safety threshold" for judges' scores.

Example: If a contestant's judge score is below a certain percentile (e.g., bottom 10% of all scores that week), they cannot be saved by fan votes alone—must be in bottom two.

Fairness rationale: Prevents extremely poor technical performers from advancing purely on popularity.

4. Judge-Fan Differential Scoring

Idea: Reward contestants who receive **similar rankings from judges and fans**, penalizing large disparities.

Example:

Final Score = Judge% + Fan% - |Judge Rank - Fan Rank|
 Final Score = Judge% + Fan% - |Judge Rank - Fan Rank|

Fairness rationale: Encourages alignment between quality and popularity, reducing "controversial" outcomes.

Query3: I need to evaluate the impact of professional dancers on contestant performance but am unsure how to quantify "dancer quality." What suggestions do you have?

Output: Quantifying dancer quality is challenging but essential. Here are some possible quantification methods:

Historical performance metrics: Calculate the average ranking, average judge scores, or proportion of final appearances for contestants previously partnered with each dancer.

Experience metrics: Total seasons the dancer has participated in, total number of contestants partnered with, years of experience on the show.

Professional background metrics: Professional achievements before joining the show, such as awards in professional dance competitions, years of teaching experience.

Compatibility metrics: The match between the dancer and the current contestant, e.g., age difference, height difference, dance style compatibility (if data allows).

In the model, you can include these metrics as independent variables or construct a composite score (e.g., through principal component analysis to reduce dimensionality). Also, consider interaction effects between dancer and contestant characteristics, as a good dancer may be more helpful for certain types of contestants.

Additionally, account for dancer fixed effects: some dancers may consistently perform well, which can be controlled by including dancer dummy variables in the model. If the data is panel-structured, you can also use a multilevel model (contestants nested within dancers) to more precisely estimate the dancer's contribution.

Finally, conduct robustness checks: try different quantification methods and observe if the results are consistent.